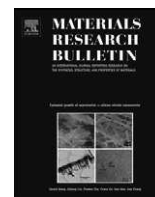




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journal homepage: www.elsevier.com/locate/matresbuSynthesis of *c*-axis oriented AlN thin films on different substrates: A review

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ABSTRACT

Highly *c*-axis oriented AlN thin films have been deposited by reactive sputtering on different substrates. The crystallographic properties of layered film structures consisting of a piezoelectric layer, aluminum nitride (AlN), synthesized on a variety of substrates, have been examined. Aluminum nitride thin films have been deposited by reactive pulsed-DC magnetron sputtering using an aluminum target in an Ar/N₂ gas mixture. The influence of the most critical deposition parameters on the AlN thin film crystallography has been investigated by means of X-ray diffraction (XRD) analysis of the rocking curve Full-Width at Half Maximum (FWHM) of the AlN-(0 0 0 2) peak. The relationship between the substrate, the synthesis parameters and the crystallographic orientation of the AlN thin films is discussed. A guide is provided showing how to optimize these conditions to obtain highly *c*-axis oriented AlN thin films on substrates of different nature.

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1. Introduction

Aluminum nitride (AlN) is an interesting III–V compound with many remarkable properties such as a wide bandgap (6.2 eV), a resistivity of $1 \times 10^{16} \Omega \text{ cm}$, and a thermal conductivity as high as 2 W/C/cm [1,2]. AlN also has satisfactory electro-acoustic properties such as moderate stiffness, as well as piezoelectric and dielectric tensors, compatibility with IC planar processing, chemical stability, etc. AlN is also a potential candidate for use in the fabrication of blue-violet light emitting diodes, short wavelength lasers, and ultraviolet light detectors. Further, the high phase velocity of acoustic waves in AlN opens up the possibility to fabricate SAW devices for high frequency applications [3,4].

Specifically, *c*-axis oriented AlN has a high acoustic velocity (12.350 m/s and 6.000 m/s for bulk and surface waves, respectively) [5], moderate electromechanical coupling coefficient kt^2 (0.8%) [6] and coupling constant K^2 (6.5%) [7], low leakage currents, as well as relatively high material *Q* of around 3000 [8]. In addition, highly *c*-axis oriented films are grown with ease on a variety of substrates [9–12] and in a wide process window, which makes them very suitable for the fabrication of thin film acoustic devices.

Layered structures of piezoelectric thin films grown on high-velocity substrates, e.g. zinc oxide (ZnO) on sapphire [13] and AlN on diamond [14,15] have already been developed, obtaining high velocities of 5.500 m/s and 6.700 m/s, respectively. For these applications as well as for other such as piezoelectric RF-MEMS, etc., it is critical to control the crystallographic orientation of the AlN

thin films. The coupling coefficient of a thin film based electro-acoustic device is closely related to the degree of *c*-axis orientation of the piezoelectric thin film. It is well known that AlN thin films can be grown epitaxially by chemical vapor deposition or molecular beam epitaxy (MBE) [16]. Under special conditions, epitaxial growth of AlN has even been achieved on silicon substrates by reactive sputtering [17]. However, low substrate temperature during deposition is required to minimize grain size as well as to decrease the process thermal budget. Some researchers have discussed the influence of deposition parameters on the orientation of the AlN films, in an attempt to find a general guideline promoting a well *c*-axis oriented thin film. The relationship between pressure and orientation is clear: the decrease in sputtering pressure results in a change of the preferred orientation of the AlN film from (1 0 0) to (0 0 0 2) [18–22] and in a decrease of the FWHM of (0 0 0 2) rocking curve [23]. However, the influence of other deposition parameters on the film quality is still under discussion. In some publications, by increasing the nitrogen concentration, the crystal quality improves [18,19,24]. However, our own experience as well as other studies [25] indicate that this is not a general tendency, and (0 0 0 2) preferred orientations can be obtained in a deposition process with a higher concentration of argon than nitrogen. Something similar can be noticed with the influence of target power (also referred to in the literature as “discharge power”); on one hand, increasing the power the crystalline quality of the AlN layers improves [18,26]. On the other hand, other publications do not agree with this conclusion, and show that reducing the discharge power, a narrower FWHM of the (0 0 0 2)-rocking curve is obtained [19,21,23], with a minimum at which the quality deteriorates again. Another example of a parameter where the tendency is not clear is the substrate temperature [19,24,26].

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Thus, the amount of parameters involved in the reactive sputtering process as well as the increasing amount of data available in the literature may become confusing or even misleading. This work aims to be a guideline for the synthesis of highly textured AlN thin films by reactive sputtering.

2. Experimental

Thin AlN films have been deposited in a home built balanced magnetron sputter deposition system. The substrates used were cleaned using a standard procedure; rinsing in pirrolidone for 5 min at 80 °C, acetone for 5 min at 60 °C and methanol in an ultrasonic bath. A lock chamber was used to load the samples into the process chamber, which has a net volume of 20 l. Prior to each run, the system was evacuated by a 800 l/s turbomolecular pump at a base pressure $<5 \times 10^{-8}$ mbar. If not otherwise indicated, the experiments were done with the substrate at room temperature [27], and this parameter was monitored with a thermocouple located 2 mm under the substrate holder. The process gases, nitrogen (N₂) and argon (Ar) were introduced into the chamber via 100 sccm/200 sccm mass flow controllers, for Ar/N₂, respectively. The gases used in this set-up were of N70 quality or better. The plasma potential was not monitored. The Al target, of 99.999% purity, was water cooled on the backside. Prior to each deposition, the Al target was conditioned for 5 min in pure Ar (Ar-flow of 30 sccm), and additional 5 min in a reactive atmosphere. The target was powered by an ENI RPG50 asymmetric bipolar-pulsed-DC generator. The applied frequency was 250 kHz while the duration of the positive pulse (pulse width) was 496 ns (13% of the total signal period), with an amplitude of +40 V.

If not otherwise specified, the parameters used on each experiment series are as shown in Table 1.

The film orientation has been determined by rocking curve measurements using a Phillips X-Pert Pro MRD diffractometer. Rocking Curve X-ray diffraction analysis is one of the most widely used techniques in crystallography. In a rocking curve measurement, the sample is rotated (rocked) through an angular range, bringing the plane in and out of the Bragg condition. The width of the measured peak, normally measured in terms of the Full-Width at Half Maximum (FWHM) value contains information of the amount by which the measured plane is off the surface normal, sometimes referred to as the “degree of orientation” of the specimen. Since the film orientation also depends on its thickness, this parameter was measured on each sample using a KLA-Tencor profilometer. For proper comparison, unless otherwise specified, only films of a thickness of 2 μm were evaluated.

3. Results

The aim of this work is to provide some understanding on how to choose the best deposition conditions to grow highly *c*-axis oriented AlN thin films on substrates of any kind (provided the substrate surface is smooth enough to allow such growth).

The most critical parameters normally considered to play a major role in the degree of *c*-axis orientation of AlN thin films deposited by reactive sputtering are the following ones:

Table 1
“Standard” process parameters.

Parameter	Value
Power [W]	900
Substrate bias [V]	No bias
Pressure [mTorr]	2
Gas composition Ar/N ₂ [sccm]	20/45
Substrate temperature [°C]	Below 80

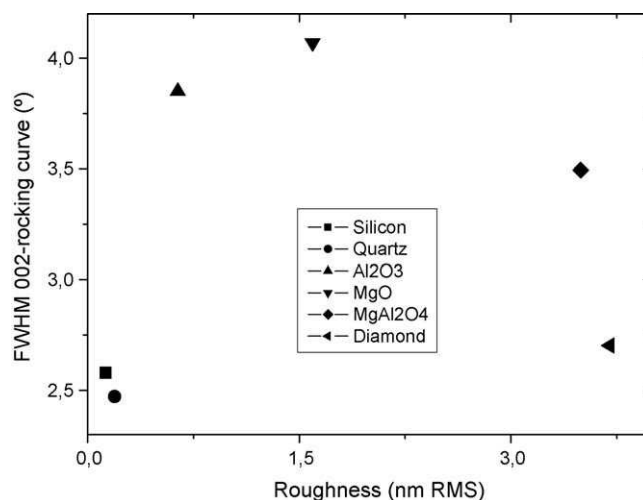


Fig. 1. Rocking curve FWHM values of *c*-axis oriented AlN thin films deposited on substrates of different material and surface roughness.

Substrate
Layer thickness
Substrate temperature
Substrate bias
Base pressure
Target–substrate distance
Deposition rate
Gas flow ratio (Ar/N₂)
Discharge power
Process pressure

We have independently varied different aspects in each of these parameters. The results of our systematic study are shown in the following:

3.1. Substrate

It is well known that a good lattice match between the substrate and the layer synthesized on top of it is an important crystal growth promoting factor [28]. The majority of the commonly used substrates have a large lattice mismatch with AlN, with the exception of SiC [29]. For instance, the lattice mismatch between AlN (0 0 0 1) and Si (1 0 0) is 42.7% [30]. In addition to high defect density, residual stress and/or a poor *c*-axis orientation may also arise from the large difference in lattice parameters. However, the lattice mismatch is not the main substrate characteristic influencing the growth of AlN thin films. Fig. 1 visualizes rocking curve FWHM values of *c*-axis oriented AlN thin films deposited on substrates of different nature (different material, different surface roughness as well as different crystallographic properties). It can be observed how, as long as the substrate surface roughness is smooth enough (below 4 nm rms¹), highly *c*-axis oriented AlN thin films can be grown on all these substrates, independently of the lattice mismatch between substrate and deposited layer.

As described in the following, other aspects such as a high surface roughness or, in the case of polycrystalline substrate materials, the grain size, have a tremendous influence on the degree of *c*-axis orientation that can be obtained. Fig. 2 visualizes the impact of the substrate grain size on the growth of AlN thin films as measured by analyzing the rocking curve FWHM values of the (0 0 0 2)-AlN peak deposited on two diamond substrates showing a similar surface roughness (3.70 nm rms and 5.85 nm

¹ “rms” stands for root mean square.

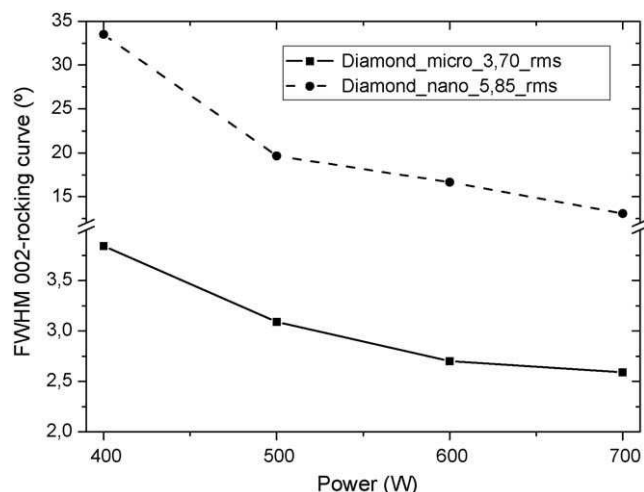


Fig. 2. In the graph, the influence of the substrate grain size on the growth of AlN thin films as measured by analyzing the rocking curve FWHM values of the (0 0 0 2)-AlN peak deposited on two diamond substrates showing a similar surface roughness, 3.70 nm rms and 5.85 nm rms, but a different grain sized, microcrystalline (Diamond_3.70_rms) and nanocrystalline (Diamond_5.85_rms).

rms, respectively) but a different grain size, microcrystalline (Diamond_3.70_rms) and nanocrystalline (Diamond_5.85_rms). Clearly, the AlN thin film grows with a much better degree of *c*-axis orientation when the substrate has a microcrystalline grain size. This grain size of the substrate seems to be a better “buffer” for the growth of polycrystalline AlN thin films with grain sizes also in the micrometer range. The graph shows also how both on micro-

well as on nanocrystalline substrates, increasing the discharge power enhances the growth in the (0 0 0 2)-direction.

Fig. 3 shows our AFM (Atomic Force Microscopy) roughness measurements of the surface of different diamond substrates. The upper part in Fig. 3 shows a 3D and height AFM (Atomic Force Microscopy) micrographs (2.5 μm size) of microcrystalline diamond (roughness of 1.907 nm rms and data scale of 17.82 nm for the 3D image). The lower part in the figure shows the results obtained for nanocrystalline diamond (roughness of 10.77 nm rms and data scale of 49.5 nm for the 3D image). In the graph in Fig. 4 the influence of the substrate roughness on the growth of AlN thin films can be observed. It is obtained by analyzing the rocking curve FWHM values of the AlN (0 0 0 2)-peak. In this case, the piezoelectric AlN layer has been deposited on two different diamond substrates with root mean square (rms) surface roughness values of 3.70 nm rms and 10.70 nm rms, respectively. As stated earlier, as long as the substrate roughness shows rms values below 4 nm, highly *c*-axis oriented AlN thin films are synthesized. On the contrary, when the substrate presents a rough surface, the growth in the *c*-axis direction is unlikely.

3.2. Layer thickness

The AlN layer thickness influences the FWHM value of the (0 0 0 2)-peak rocking curve. As visualized in Fig. 5 the degree of *c*-axis orientation depends on the film thickness up to a value of approximately 1 μm . Once this value is reached, there is no influence of the layer thickness on the FWHM value obtained in our X-ray diffraction analysis. A similar effect has also been observed in [31]. It should be noticed that this is valid for three substrates belonging to the three different crystallographic structures;

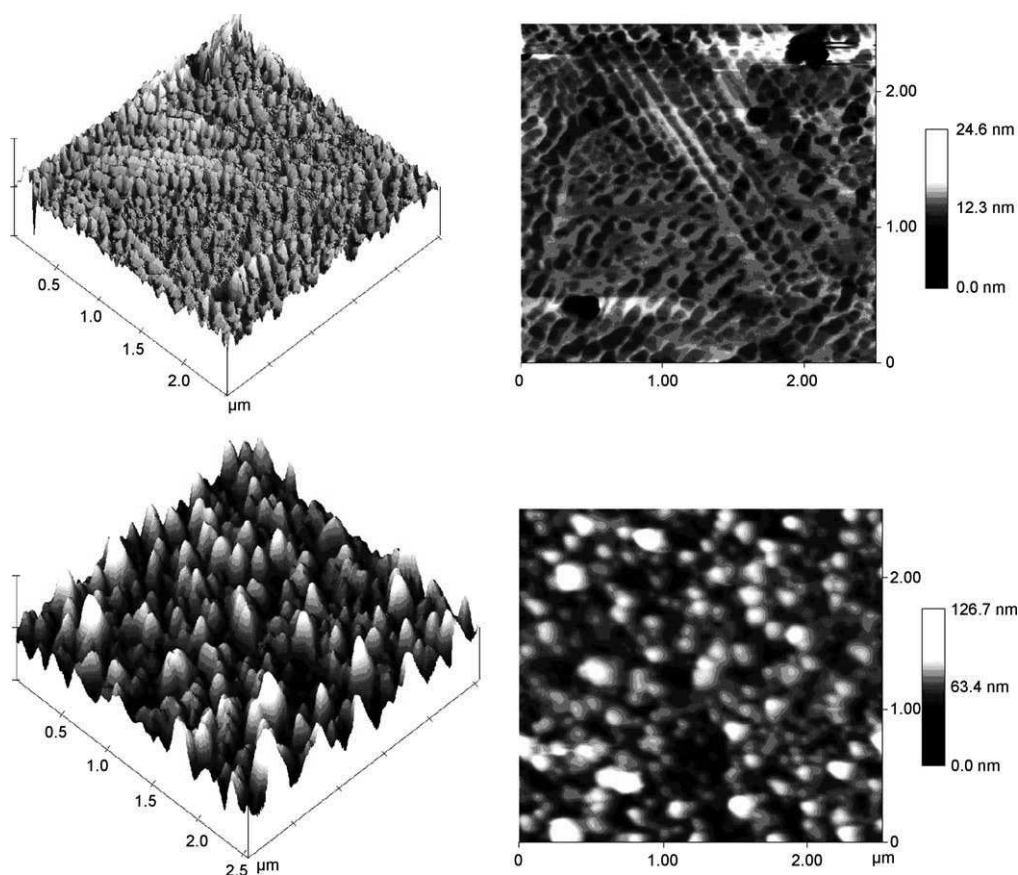


Fig. 3. 3D and height AFM micrographs (2.5 μm size) of (i) microcrystalline diamond (roughness of 1.907 nm rms, and data scale of 17.82 nm for the 3D image), top, and (ii) nanocrystalline diamond (roughness of 10.77 nm rms, and data scale of 49.5 nm for the 3D image) bottom.

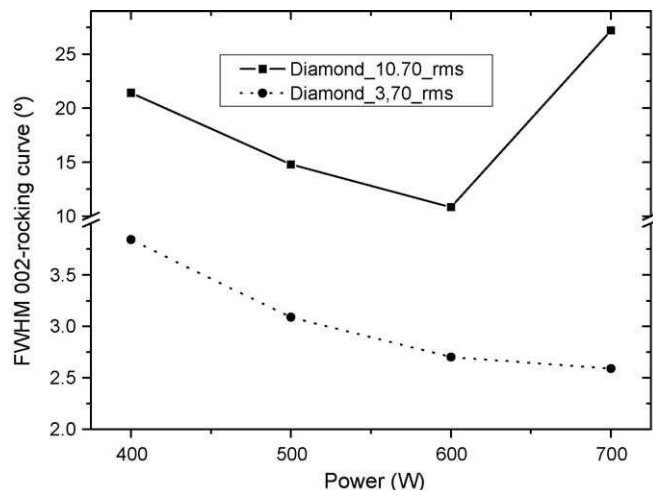


Fig. 4. In the graph, the influence of the substrate roughness on the growth of AlN thin films as measured by analyzing the rocking curve FWHM values of the (0 0 2)-AlN peak deposited on two diamond substrates of very different surface roughness, 3.70 nm rms and 10.70 nm rms, respectively.

monocrystalline (silicon), polycrystalline (synthetic diamond) and amorphous (quartz).

3.3. Substrate temperature

Epitaxial AlN films grown by chemical vapour deposition (CVD) on sapphire have been used to fabricate high frequency surface acoustic wave resonators and filters with near theoretical performance. However, the required substrate temperature (over 1000 °C) is high and makes the process cost ineffective and incompatible with semiconductor IC processing. Low-temperature growth of AlN thin films on silicon (Si) substrates is also very important for the development of new hybrid optical integrated circuitry (OIC). As shown in Fig. 6 when using optimized parameters in the synthesis of AlN thin films by reactive sputtering, the use of substrate temperature to promote the growth in the (0 0 2) direction is not strictly necessary.

3.4. Substrate bias voltage

The effect of substrate bias voltage during sputtering is known to enhance the kinetic energy of the positive ions in the plasma,

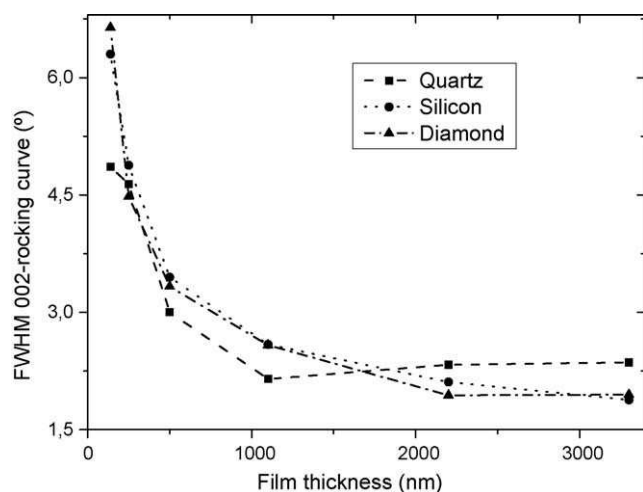


Fig. 5. Rocking curve FWHM values of AlN layers as a function of the AlN film thickness.

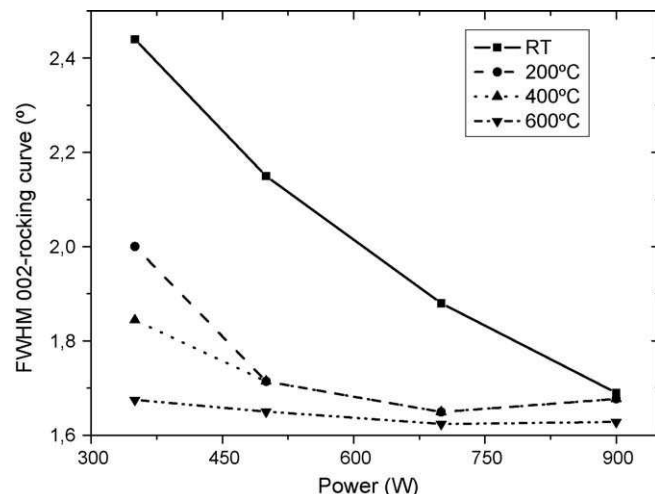


Fig. 6. Rocking curve FWHM values of highly (0 0 2) oriented AlN thin films deposited at different substrate temperatures.

attract them towards the substrate and thus change the microstructure of the deposited thin films. At low processing pressures (below 3 mTorr, although this is also system dependent) highly oriented AlN thin films can be obtained without substrate bias. As visualized in Fig. 7 a small substrate bias voltage (around 10–20 V) may slightly improve the degree of *c*-axis orientation. However, if a higher substrate bias is applied, stress will be introduced at the film–substrate interface [32] and other orientations such as (1 0 0) or (1 0 1) may appear [33]. This will result in a diminished coupling coefficient [34] of an electro-acoustic device fabricated using this material.

3.5. Base pressure

The base pressure (pressure inside the vacuum chamber prior to the process start) is also known to be a very critical parameter. Base pressures below 10^{-7} mbar are necessary to prevent the incorporation of oxygen into the growing AlN thin film. It is well known that Al has a strong chemical affinity to oxygen for the formation of Al_2O_3 . The formation of Al_2O_3 in the AlN films will retard the single grain columnar growth, and leads to a poor preferred orientation as well as poor surface topography. An oxygen content in the thin film as small as 5% will have a detrimental effect on the degree of *c*-axis orientation.

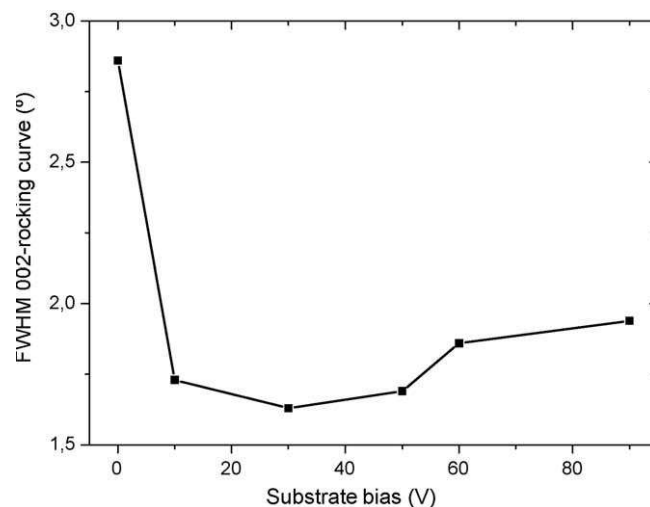


Fig. 7. Rocking curve FWHM values showing the effect of substrate bias on highly *c*-axis oriented AlN thin films.

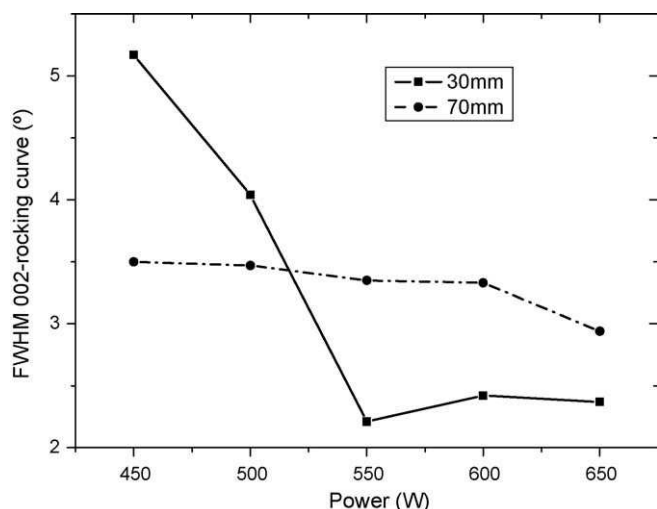


Fig. 8. Rocking curve FWHM values showing the degree of orientation of AlN deposited by reactive sputtering at two different target–substrate distances.

3.6. Target–substrate distance

Fig. 8 shows the rocking curve FWHM values of the (0 0 0 2)-peak of AlN thin films deposited using two different target–substrate distances, namely 30 mm and 70 mm. Using a low discharge power, a slight improvement is obtained by increasing the target–substrate distance to 70 mm. However, better results are obtained at shorter distances when using higher target powers. Other target–substrate distances such as 45 mm have also been investigated, obtaining similar FWHM values. These results, as well as the high amount of data available in the literature, showing highly textured AlN thin films at a multitude of target–substrate distance values [35–37], lead us to the conclusion that this parameter by itself does not play a significant role. In other words, highly oriented AlN thin films can be obtained in a wide range of target–substrate distances going from 30 mm to 70 mm and above.

3.7. Deposition rate

The rate at which the AlN thin film is deposited on the substrate is mainly related to the discharge power, to a lesser extent to the process pressure and slightly to the Argon content in the Ar/N₂ mixture. All these parameters are discussed independently in this work. A relatively high deposition rate reduces the impurity incorporation into the film from background gases. On the other hand, high deposition rates result in a decrease in the ion/atom flux ratio at the substrate during ion assisted deposition resulting in worsened texture as well as tensile stress. Finally, extremely high deposition rates may result in non-stoichiometric films as well as structural defects due to reaction and diffusion limitations. Low deposition rates on the other hand result in poor film quality mainly due to background gas incorporation. The deposition rate for a successful synthesis of reactive sputtered AlN thin films oriented in the (0 0 0 2) direction should be between 15 nm/min and 35 nm/min.

3.8. Gas flow ratio (Ar/N₂)

AlN thin films deposited in a higher nitrogen concentration are likely to exhibit a higher crystallinity and refractive index because of less nitrogen deficiency. As the sputtering pressure approached the transition pressure range, nitrogen partial flow rate determines the preferred orientation of deposited films. Fig. 9 shows the FWHM of the (0 0 0 2)-AlN peak of films (deposited using several

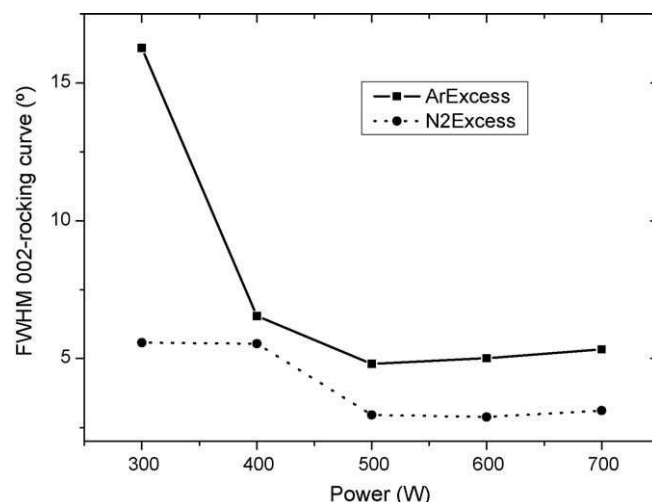


Fig. 9. Rocking curve FWHM values of highly oriented AlN films deposited under different gas flow regimes.

discharge powers) under excess of Ar and N₂. According to these results, the Ar/N₂ gas flow ratio should be kept below 1. However, having for instance ratios of Ar/N₂ = 2, the synthesis of AlN films in the (0 0 0 2) direction is also possible. Although the deposition rate will increase with the amount of Ar, thus favoring a faster processing, the difference in the deposition rate between using Ar/N₂ ratios of 1 and 2 will not be more than around 10%, as shown in Fig. 10. According to our results, an Ar/N₂ ratio of approximately 2/3 is recommended.

3.9. Discharge power

The discharge power is indeed a parameter to be optimized in order to increase the crystallographic quality. The effect of the discharge power on the orientation of the growing films typically shows a trend as the one visualized in Fig. 11. The AlN thin films deposited at low discharge power (i.e. below 300 W in this work) do not exhibit any crystalline orientation, even if the process pressure is low. It may be due to the fact that the atoms sputtered towards the substrate did not possess (as a collective) the required energy to form the well-oriented structure because the power is low. In other words, the process is outside the desired energy window to promote crystal growth [18,38].

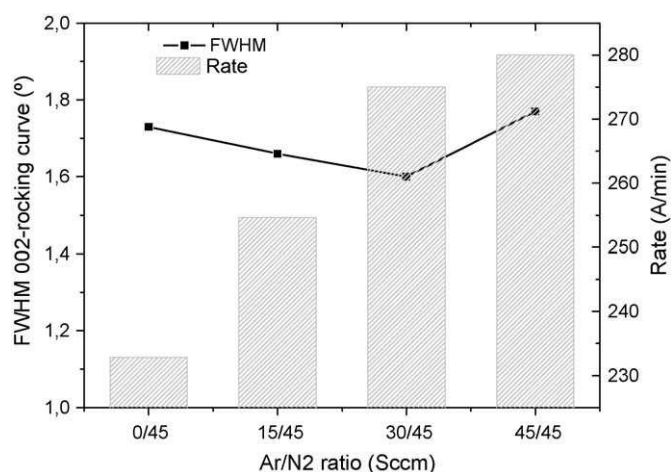


Fig. 10. Rocking curve FWHM values and deposition rate of highly c-axis oriented AlN films as a function of the Argon content.

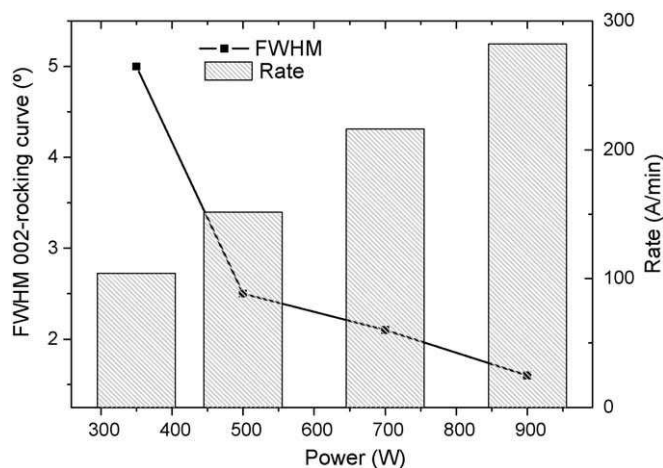


Fig. 11. Rocking curve FWHM values and deposition rate of highly oriented AlN films synthesized using different target power values.

3.10. Process pressure

Together with the discharge power, the process pressure is by far the most influencing parameter on the degree of *c*-axis orientation of AlN thin films deposited by reactive sputtering. An increase of processing pressure affects negatively the preferential growth in the (0002) direction, although it is possible to obtain highly oriented films even at pressures as high as 8 mTorr, as shown in Fig. 12. Nevertheless, it can be stated that a low pressure promotes the growth of the AlN thin film in the *c*-axis orientation. However, a pressure below 1 mTorr may cause instability in the plasma and may introduce a high amount of stress at the film-substrate interface [32,39,40]. A process pressure between 2 mTorr and 3 mTorr is recommended. The process pressure used in these experiments was 3 mTorr, but similar results have been obtained using processing pressures between 2 mTorr and 6 mTorr, as seen in Fig. 12. According to the kinetic theory of molecular gases, the mean free path of a gas molecule at constant temperature has a value that is inversely proportional to the pressure. Thus, the kinetic energy transfer from the plasma to the growing film surface is enhanced as the sputtering pressure decreases. The *c*-axis orientation, therefore, is increased due to the increase of kinetic energy transfer at the film surface as the sputtering pressure decreases. At low process pressures, the AlN layer surface may show a non-uniform grain growth, which may be due to stress-induced in the AlN thin film. On the other hand, when the

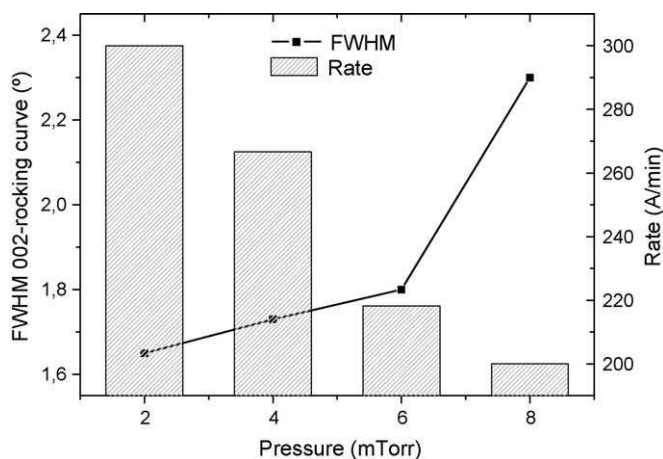


Fig. 12. Rocking curve FWHM values and deposition rate of highly oriented AlN films synthesized at different process pressures.

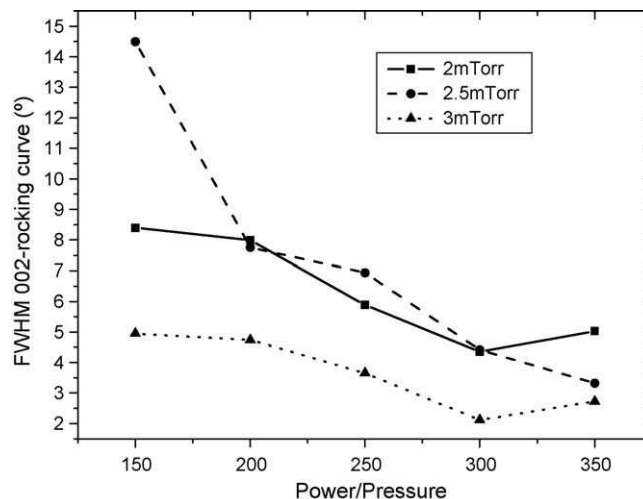


Fig. 13. Rocking curve FWHM of (0002)-AlN peak of thin films deposited under different discharge power/process pressure ratios.

sputtering pressure is increased, the surface exhibits a wave-like stripe texture, which may be due to the coexistence of some types of crystals. Further increase of the sputtering pressure, may give rise to a surface showing small grain aggregates with a closely packed texture.

Summarizing, after having investigated all deposition parameters, we conclude that two of them, namely the process pressure and the discharge power, have a strong influence on the degree of *c*-axis orientation of the AlN thin films. Fig. 13 shows the FWHM of the AlN (0002) peak for a number of films grown at different “discharge power/pressure”. Three different ratios have been investigated for pressure series at 2 mTorr, 2.5 mTorr and 3 mTorr whereas the discharge power was varied from 300 W up to 1050 W. The results show a trend in which higher power/pressure ratios promote the growth of the AlN thin films in the (0002) direction.

4. Conclusion

We conclude from this investigation that substrate temperature, substrate bias or a good lattice match between the substrate and the deposited layer are not strictly necessary to deposit highly *c*-axis oriented polycrystalline AlN thin films on substrates of different nature (different material as well as different crystallographic properties). A process pressure between 2 mTorr and 3 mTorr is recommended as well as a gas flow ratio of approximately Ar/N₂ = 1/3. Increasing the “discharge power/pressure” ratio also promotes the orientation of the films in the (0002) direction.

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